

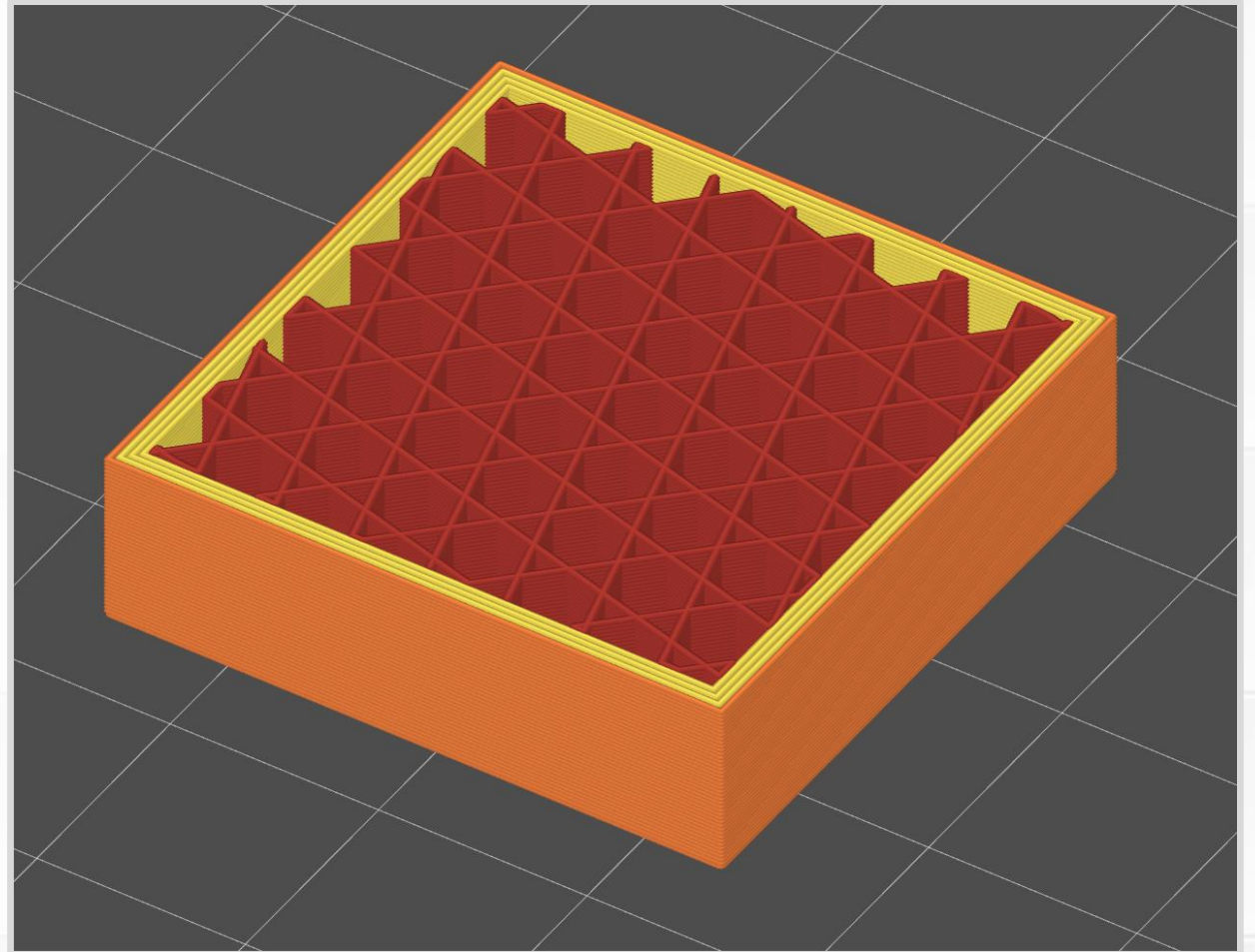


FFF Printing Fundamentals

Part 4: Advantages

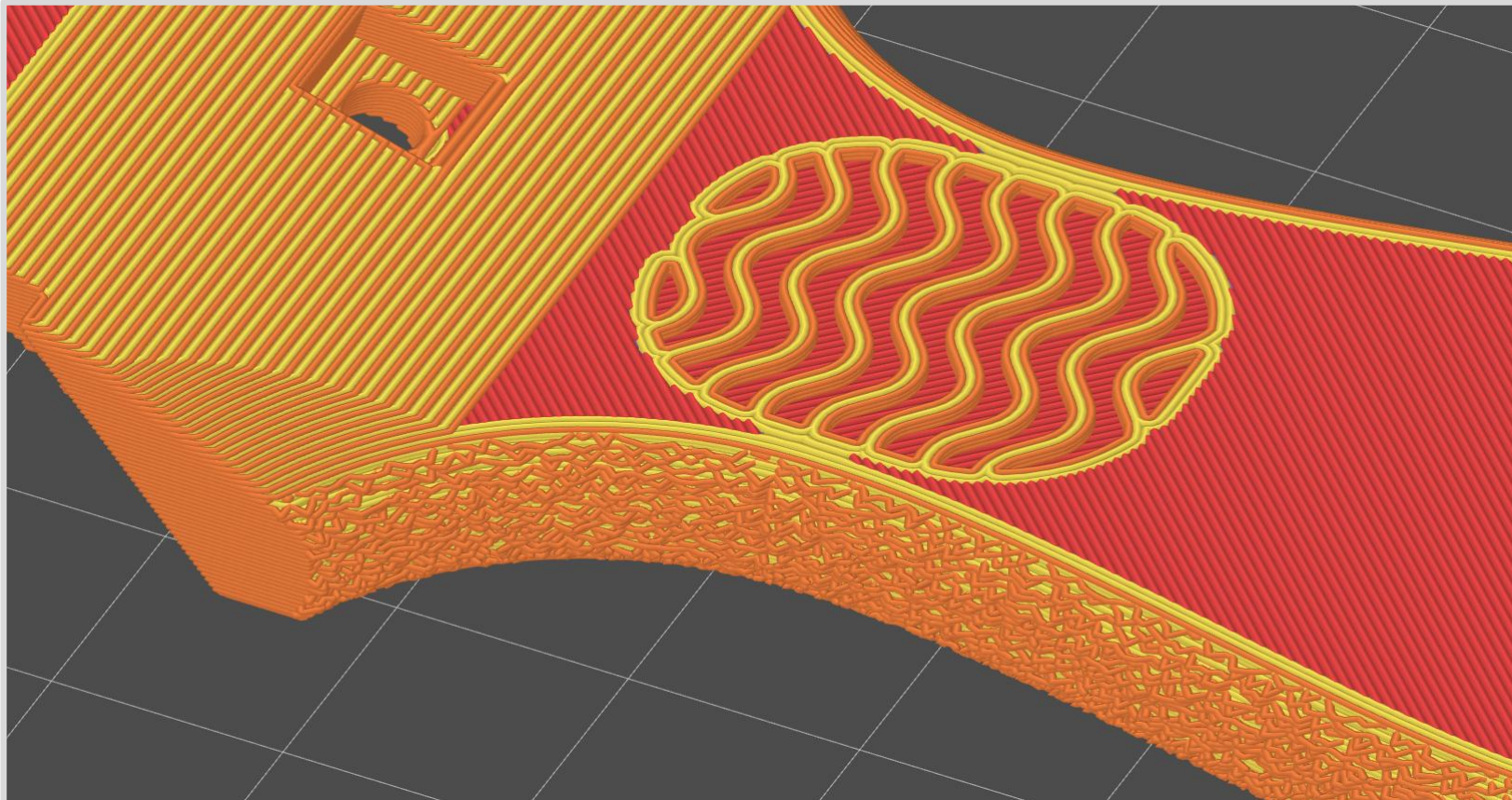
FFF Advantages

- Printers use tiny “building blocks.”
- Typical extrusions:
 - **0.2 mm tall** (1-2 pieces of paper).
 - **0.4 mm wide**.
- Not limited to uniform prints.
 - *Example:* Solid exterior and sparse infill.
- Great at making hollow objects.



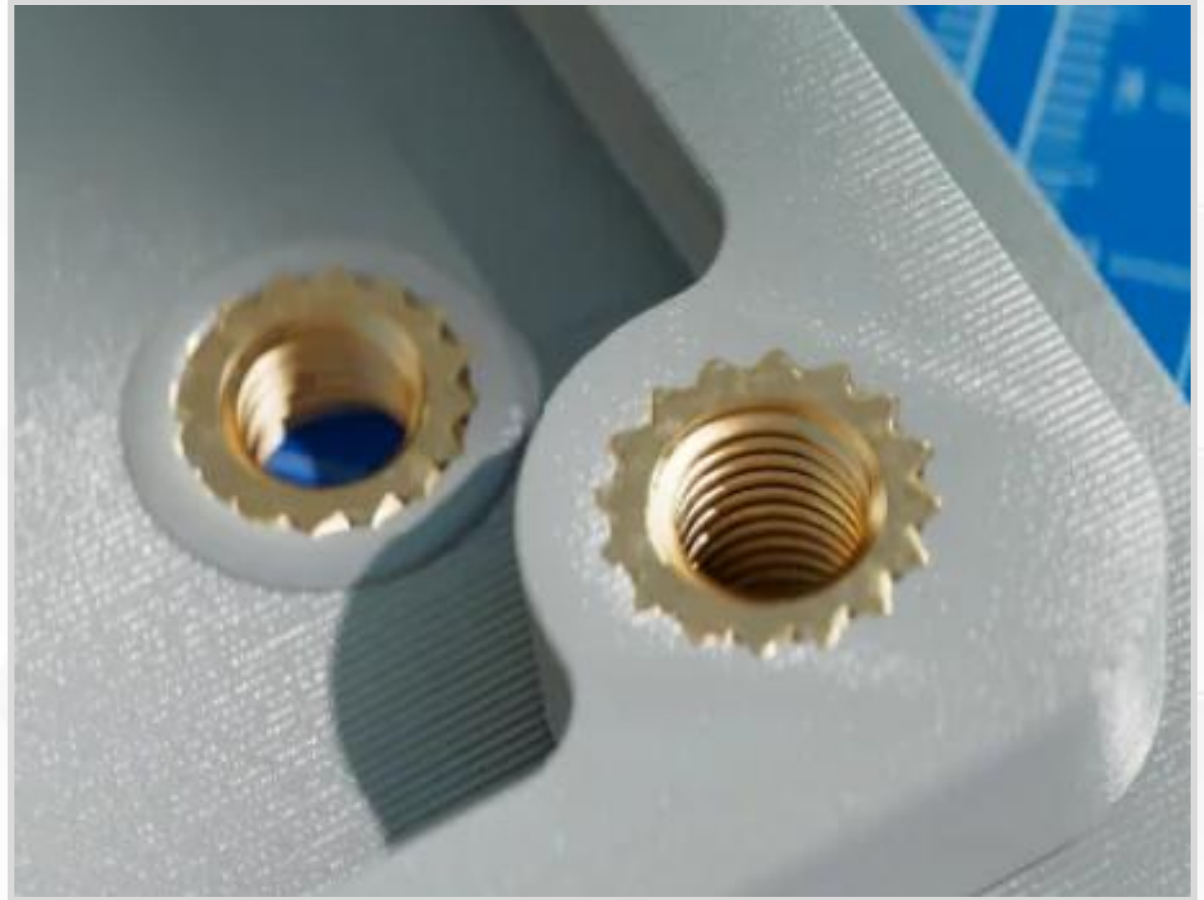
Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Plastic is soft → Screws easily wear out plastic holes.
- Threaded anchor for screws.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



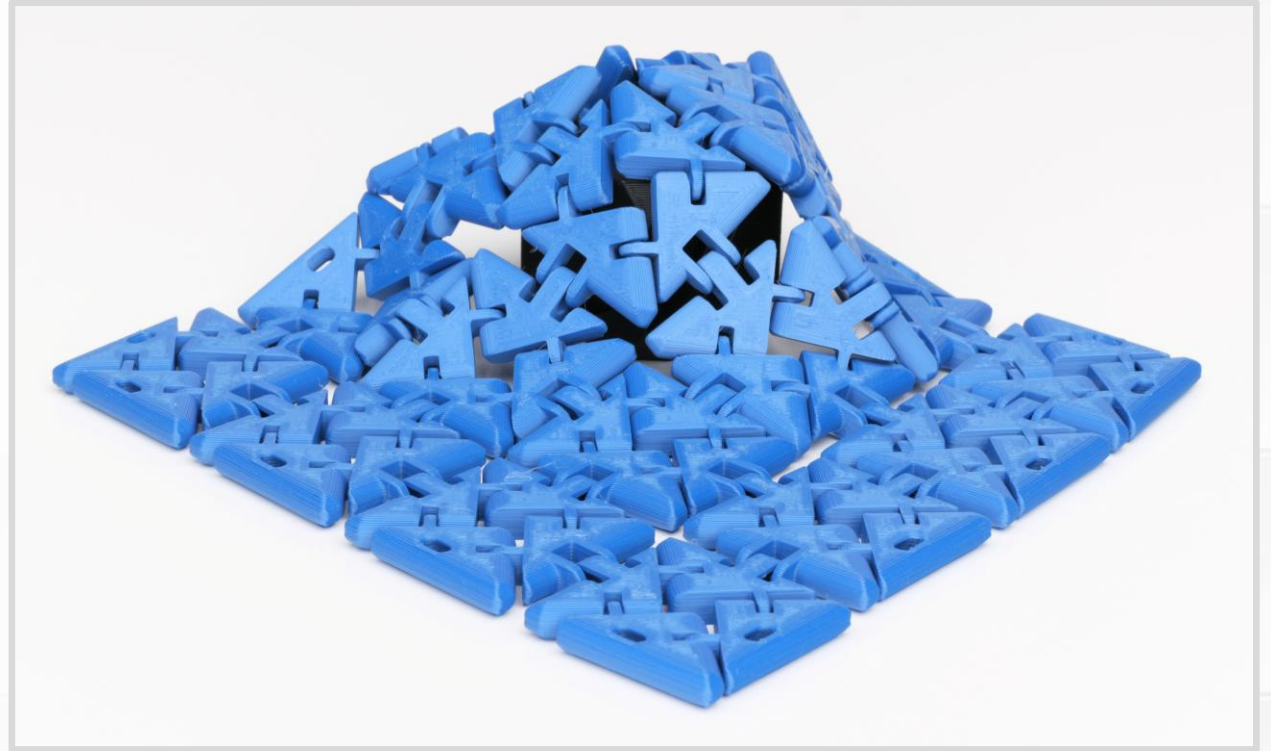
(Pictures from [CNC Kitchen](#))

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
 - Nuts and fasteners.
 - Magnets.
 - Weights.
 - Reinforcing plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
 - Need to be at the printer at the right layer.

Print-in-Place (PIP)

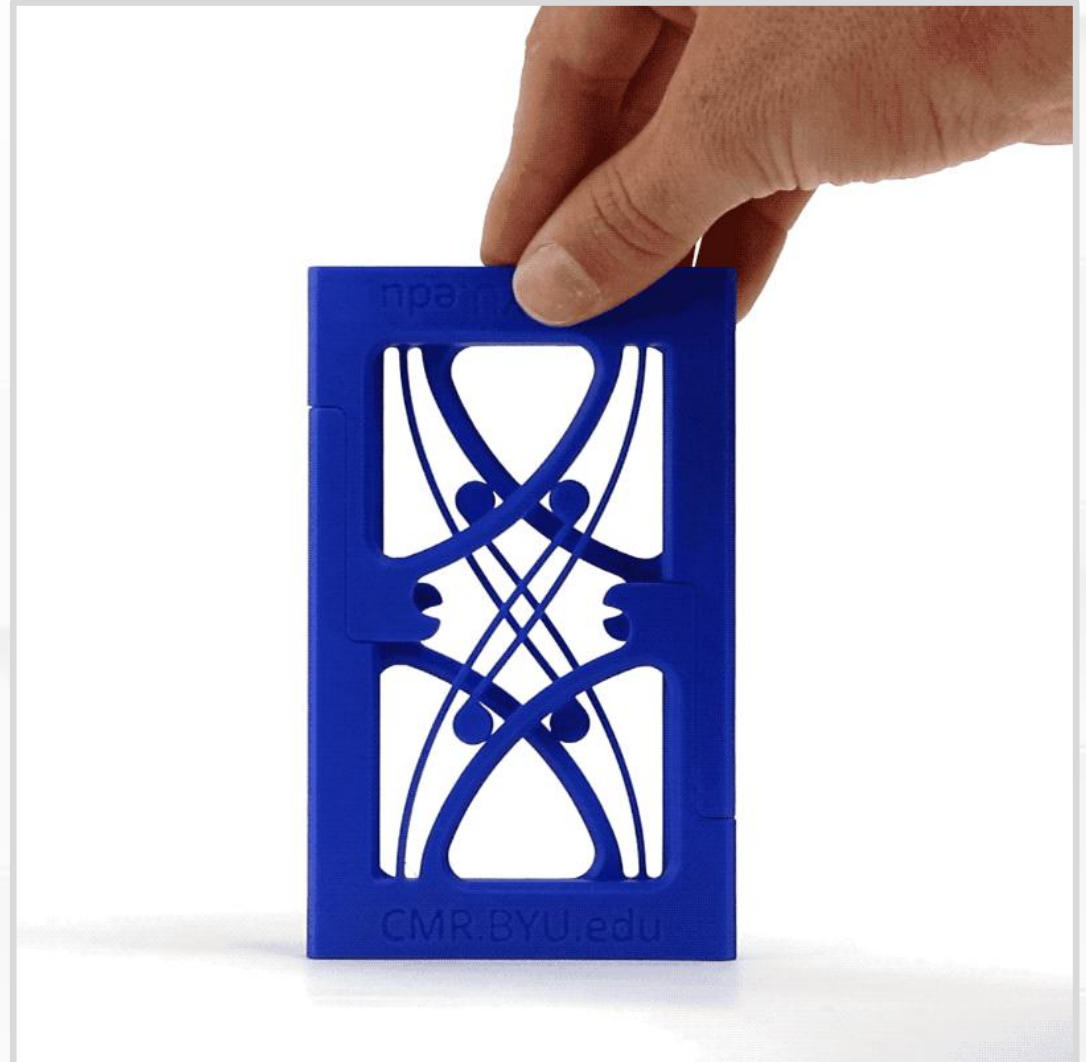
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by [printables.com/@Flying Gyroscope](https://printables.com/@FlyingGyroscope)



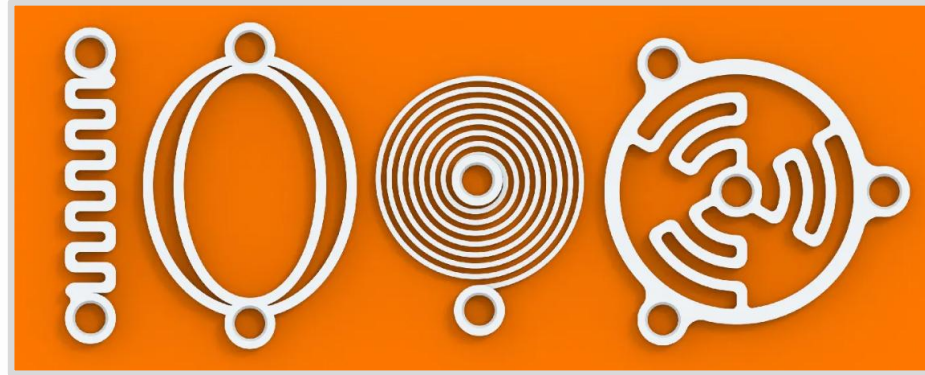
Functional Flexing

- Springs and **compliant mechanisms**.
- Features/shapes that function by bending and flexing.
- Elastic materials.
 - Springy.
 - No permanent deformation.
- Remember strength and layer lines.
 - Flex springs in XY plane for maximum durability.

(Animation, Constant Force
Mechanism by BYU CMR)



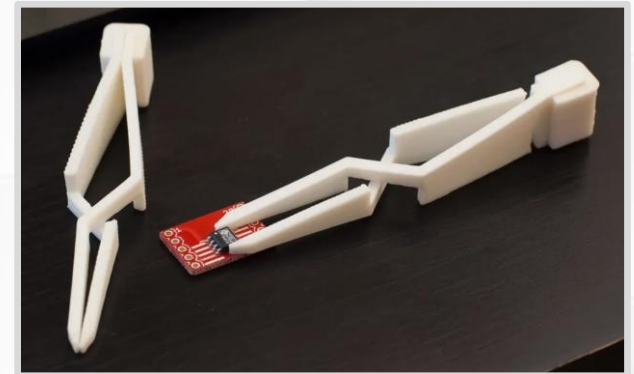
Springs by VC Design



Clip by AE



Cross tweezer by joo



Living hinge by DrJones

